Provisional figures and tables

MTU15 thesis (companion to paper.tex)

Pablo Paramio

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This file is exploratory scratch space. None of these objects is currently cited from `paper.tex`. They are kept here for inspection and decision-making — promote to the main paper if the empirical content is load-bearing, otherwise leave here or move to `attic/`.

Contents

1 IDA pre/post-blackout near-clearing dissimilarity

New empirical test (2026-05-16). For each `(firm, unit, date, hour)` cell of CCGT IDA bids, compute the kernel-weighted within-hour quarter dissimilarity D_w (Epanechnikov kernel, $h = 20$ EUR/MWh, centred on the hour-mean IDA clearing price). Compare two regimes that hold the MTU15-IDA clock constant:

- **Pre-blackout**: 2025-03-19 → 2025-04-27 (40 days; post-MTU15-IDA, reforzada-free).
- **Post-blackout**: 2025-04-28 → 2025-09-30 (155 days; reforzada active throughout).

The hypothesis being tested: under reforzada, REE forces CCGT dispatch via post-clearing intervention, so the marginal-bid leverage on the clearing price is reduced and within-hour granularity exploitation should weaken (pre > post). The data reject this prediction.

Table 1: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the ±20 EUR/MWh band around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 → 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 → 2025-09-30 (155 days, reforzada active).

	Pre-blackout			Post-blackout		
Firm	Critical	Flat	Crit−Flat	Critical	Flat	Crit−Flat
IB	9.4	7.0	+2.4	18.6	6.2	+12.4
GE	12.4	8.1	+4.3	36.1	21.4	+14.7
GN	29.6	9.2	+**20.4**	50.3	36.2	+14.1
HC	27.9	20.0	+7.9	32.3	20.3	+12.0

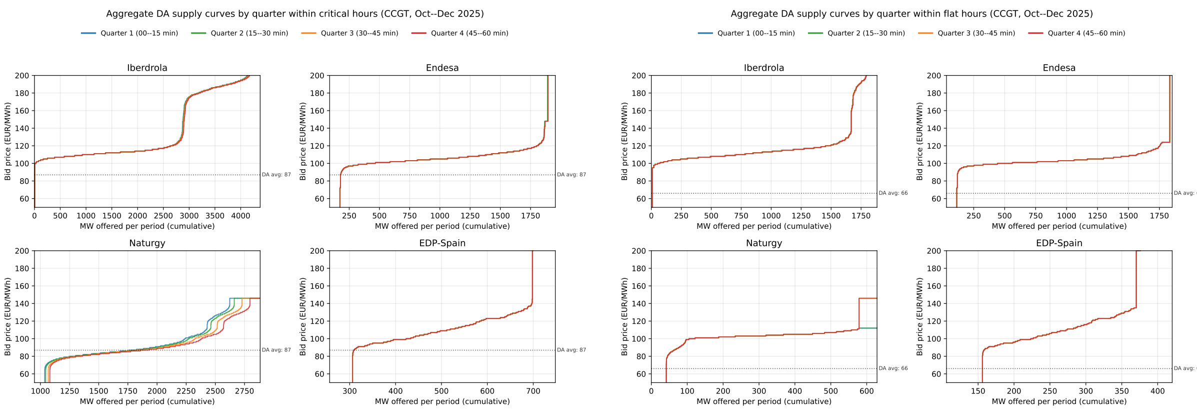
Reading. (i) Pre-blackout, only Naturgy shows a clean critical-vs-flat strategic gap; the other three are noisy. (ii) Post-blackout, all four firms show a +12 to +15 percentage-point gap — granularity exploitation broadens. (iii) The directional prediction (pre > post) is rejected for every firm. (iv) Flat-hour rates post-blackout are non-trivial (6–36%), which the kernel-weighted falsification ought to keep at zero if all variation were strategic; the elevated flat-hour baseline is itself a reforzada signature (overnight CCGT commitments creating quarter-level operational adjustments).

A coherent alternative narrative: reforzada *amplifies* the inframarginal volume on a forced-up CCGT (because REE pays Pmin out of merit at the unit's own price-of-restrictions offer), so a higher DA / IDA clearing price is worth proportionally more on the forced block. The strategic incentive to push the clearing price via within-hour bid shaping rises, not falls.

Source data. `scripts/analysis/bid/quarter_dissimilarity_ida.py`; outputs at `results/regressions/`

2 Per-quarter CCGT bid curves (DA)

The within-hour quarter dispersion is the visual companion to Table 5 of the main paper. The aggregate construction (averaging across ∼92 dates per cell) smooths day-specific variation and tends to hide the strategic shaping; the kernel-weighted dissimilarity in the main paper is the right quantitative complement.



(a) Critical hours (05:00–09:00, 16:00–23:00)   **(b)** Flat hours (01:00–04:00) — falsification

Figure 1: Aggregate DA supply curves by quarter, CCGT, Oct–Dec 2025. Aggregated across ∼92 dates per (firm, hour, quarter) cell; day-specific quarter dispersion is smoothed in this visual.

3 Per-firm CCGT bid curves and per-quarter pairs (IDA)

The IDA analogues of the main-text DA figures. The IDA empirical pattern in the post-MTU15-DA window matches the DA pattern: dispersion is concentrated in Naturgy and visible only in critical hours.

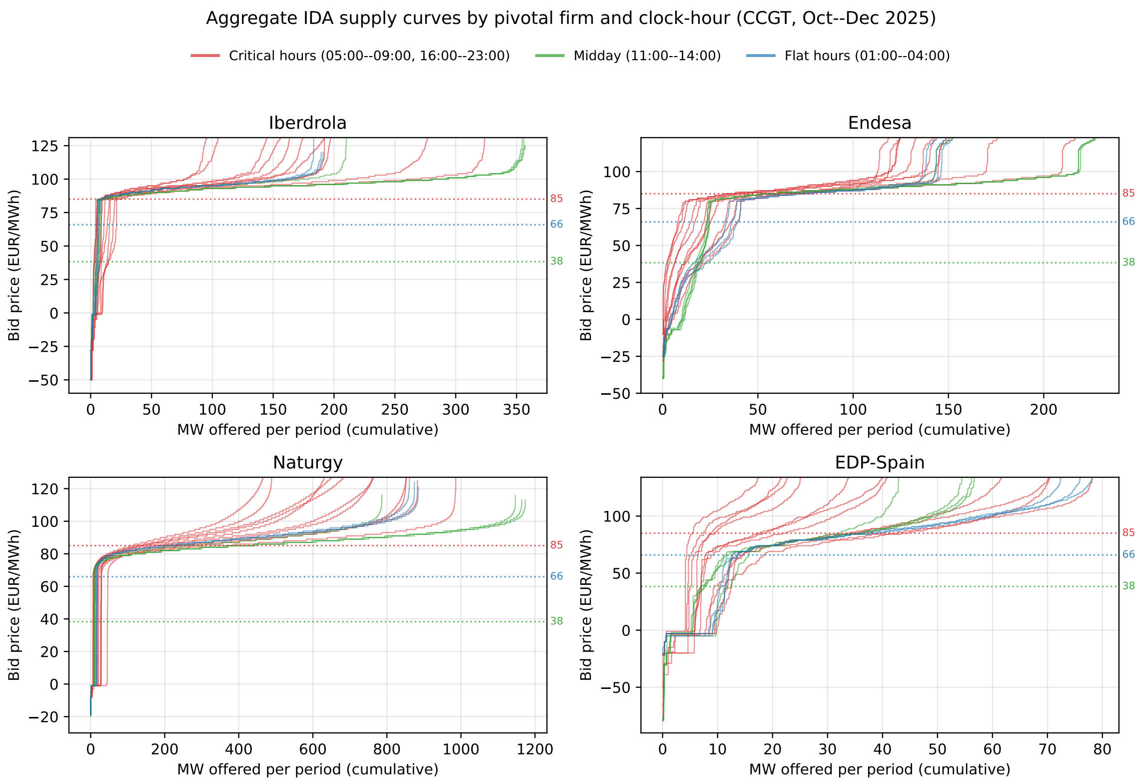
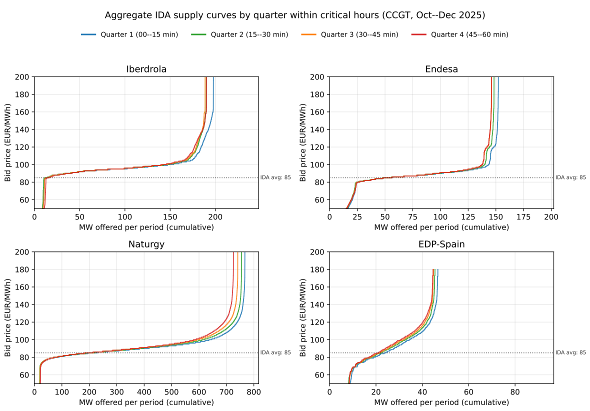
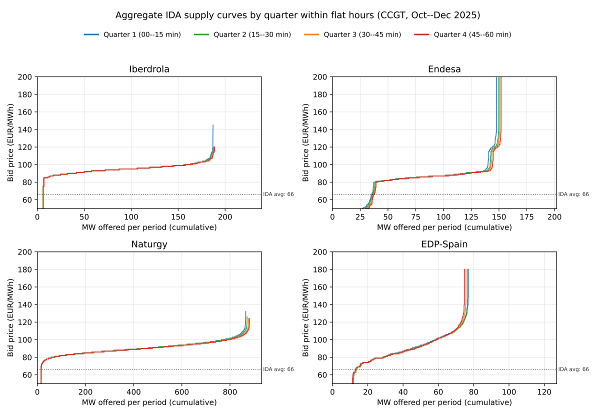


Figure 2: Aggregate IDA supply curves by pivotal firm and clock-hour, CCGT, Oct–Dec 2025.



(a) Critical hours (05:00–09:00, 16:00–23:00)

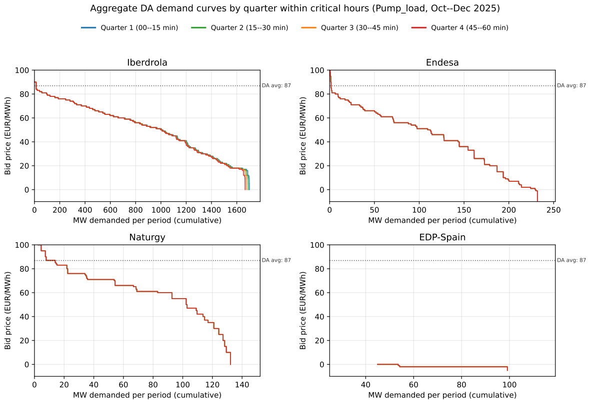


(b) Flat hours (01:00–04:00)

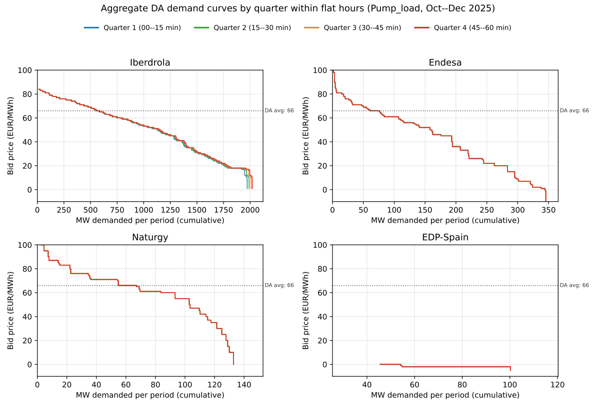
Figure 3: Aggregate IDA supply curves by quarter, CCGT, Oct–Dec 2025.

4 Per-quarter pump-storage demand curves (DA + IDA)

Per the main-text Reading, pump-storage operators do not differentiate charge bids across the 4 quarters of a clock-hour — the within-hour granularity is unused on this margin. Visuals retained here in case future scrutiny needs them.



(a) DA, critical hours



(b) DA, flat hours

Figure 4: Aggregate DA demand curves by quarter, pump-storage charge bids, Oct–Dec 2025.

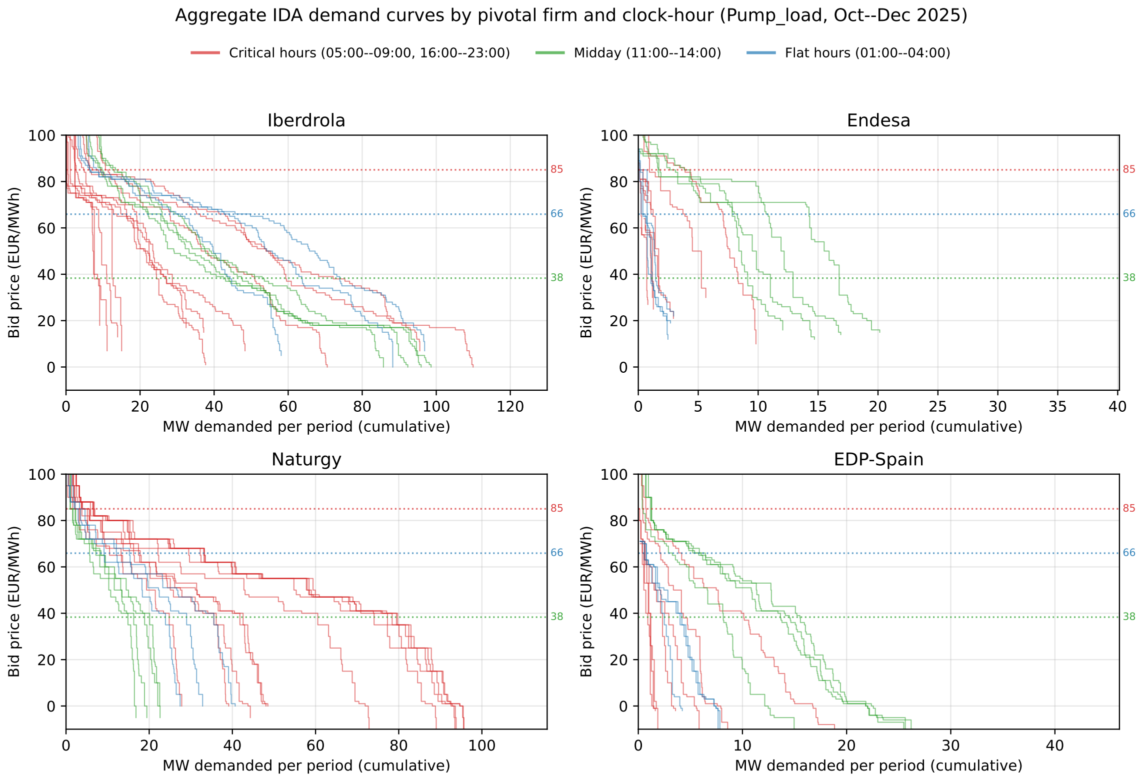
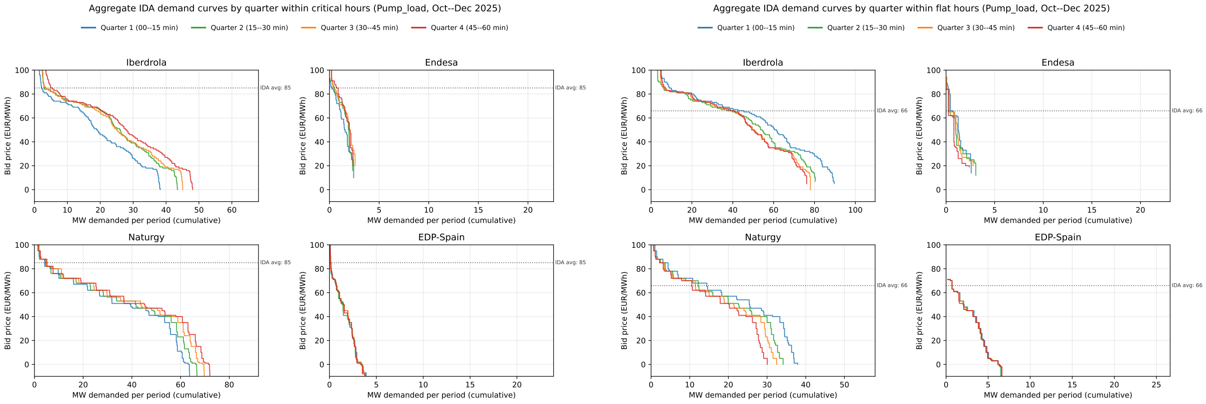


Figure 5: Aggregate IDA demand curves by pivotal firm and clock-hour, pump-storage charge bids, Oct–Dec 2025.



(a) IDA, critical hours

(b) IDA, flat hours

Figure 6: Aggregate IDA demand curves by quarter, pump-storage charge bids, Oct–Dec 2025.

5 Operational vs strategic decomposition (provisional)

The original main-paper Table 6 reported a critical–flat asymmetry of the within-hour bid-curve dissimilarity flag, with the interpretation "only Naturgy CCGT shows the strategic signature". The IDA pre/post-blackout analysis above complicates this: under reforzada, all four firms show the strategic signature in IDA, not just Naturgy. The current main-paper framing is regime-specific to the DA Oct–Dec 2025 window. Until we have a unified read, the table is held here.

Table 2: Original DA Oct–Dec 2025 critical–flat asymmetry table. Naturgy CCGT shows the strategic signature; renewables and retailer demand vary symmetrically across hour-classes (consistent with operational refinement); pump-storage is mechanical. Compare with the IDA pre/post-blackout finding in §1.

Channel	Tech	Firm	Critical	Flat	Crit − Flat
Strategic	CCGT	GN	24.5	0.0	+24.5
Strategic	CCGT	IB	5.8	3.9	+1.9
Operational	Wind	IB	97.2	97.3	−0.1
Operational	Wind	GN	92.7	92.6	+0.1
Operational	Wind	HC	93.7	93.8	−0.1
Operational	Solar PV	GN	15.9	0.0	+15.9
Operational	Retailer	GE	76.5	76.3	+0.2
Operational	Retailer	IB	40.9	40.2	+0.7
Operational	Direct_consumer	IB	96.0	95.3	+0.7
Mechanical	Pump_load	IB	3.2	5.5	−2.3
Mechanical	Pump_load	GE	0.0	0.0	0.0
Mechanical	Hydro_pump	IB	0.4	0.0	+0.4
Mechanical	Hydro_pump	GE	0.0	0.0	0.0